

ANDROID DATA EXTRACTION TOOLS

COMMERCIAL VS OPEN SOURCE

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1 Executive Summary

This paper compares and contrasts two different methods of extracting data in Android. Magnet AXIOM's commercial forensic solution that employs ADB Unlocked acquisition technique, and a manual method based on Android Debug Bridge commands. This paper aims at comparing and contrasting the effectiveness, limitations as well as the role of both techniques in digital investigation. For the purpose of this comparison, a practical test was carried out on a Samsung SM-N9005 Android 5.0 with root permissions granted. The two methods were compared according to the number and types of forensic artifacts that can be obtained, the depth of access, usability, and forensics.

Performing a full file system acquisition with Magnet AXIOM using ADB Unlocked, the data types that were recovered included web browser history from Chrome, Firefox, and Samsung Browser, communication data such as contacts, media in form of images and videos, documents in form of PDFs and text files, location data in form of Wi-Fi profiles and Google Maps, and system data in form of connected Bluetooth devices, operating system details, calendar entries, and emails. The simplicity of the software and the ability to parse through the information made AXIOM effective for categorising and reporting of the extracted information.

On the other hand, the manual ADB method entailed the use of adb devices as well as adb pull commands to directly download data from other directories including; /storage/emulated/ and /data/data/. Root access allowed for deeper analysis of the application specific data as most of the directories were inaccessible in non-rooted devices. This method offered a great deal of flexibility in the extraction process, but it was complex, not very automated, and did not include an artifact parsing feature.

The results show that both methods can effectively recover the forensic data, but AXIOM has more systematic and legal admissible procedure than ADB that provides more flexibility and deep level access to the root of the device. The report discusses the ethical, legal, and technical challenges involved in both methods, including concerns related to data integrity, rooting risks, and the forensic admissibility of evidence. This report highlights the complementary strengths of commercial and non-commercial Android acquisition methods and provides recommendations for their appropriate use in forensic investigations.

2 Background

2.1 Introduction to Android Forensics

With smartphones becoming an essential tool in people's daily lives, both personal and professional, Android forensics has become an important subfield of digital forensics. Android is an open platform and is used by a large number of people, which can be both a strength and a weakness for a forensic examiner. According to Almuqren et al. (2024), due to the number of different devices, manufacturers, file systems, and possible security configurations, the process of obtaining and analysing Android data is not easy.

Digital forensics on Android devices can be done on internal memory, external memory such as SD cards, system partitions, and cloud backups. It can be affected by some factors like device encryption, bootloader state, root state, and the Android version (KrishnanZhou and An, 2019). This is important in making the right decision when choosing the data acquisition method to use.

2.2 Data Acquisition Process and Approaches

In mobile forensics, there are three main approaches of data collection (Kumar, 2021):

- **Logical Extraction**

This is the process of pulling data from the operating system using the standard API or sync mechanisms. It often copies files that are visible to the user but not the files that have been deleted or are hidden.

- **File System Extraction**

This approach captures the entire file system, including hidden and deleted files within directories. It allows forensic practitioners to access application data, logs, and system files that are not visible during logical acquisition. Root access or vendor-specific exploits may be required to perform this extraction (MahalikRohit and Bommisetty, 2019).

- **Physical Extraction**

It refers to a process of transferring physical memory byte by byte. Although it is the most invasive method (gaining access to deleted files and slack space), it is usually used with advanced techniques or hardware.

ADB is a tool for Android devices working in developer mode and provides interaction with the devices as well as logical or partial access to the file system (Android, 2025). When used together with root access, ADB can pull more information from restricted folders such as /data/data/. Other commercial tools such as Magnet AXIOM, Cellebrite UFED, and MSAB XRY are based on these principles but come with features such as automation, graphical user interface, parsing, and report generation that makes them ideal for professional use..

2.3 Android Security and Root Access

Android has several security features such as app sandboxing, secure boot mode, file-based encryption, and SELinux (Mayrhofer et al., 2021). These mechanisms safeguard user information and ensure that it cannot be accessed by anyone who should not be able to access it. However, forensic acquisition often requires the circumventing or even the neutralization of some of these controls.

Rooting, which is the process of gaining the root access, allows the examiner to penetrate through Android application container and gain full access to the file system (Hassan and Pantaleon, 2017). However, this process can alter key data points. For instance, rooting may modify the /system partition, introduce new binaries like, su and busybox, change access and modification timestamps, or trigger OS-level security alerts (Krishnan, 2014). Pertaining to the forensic approach, data must be as original as possible, and the process should be as consistent and non-invasive as possible. Therefore, rooting should be done prudently, especially in legal matters where the chain of evidence and alteration of evidence may arise.

2.4 Magnet AXIOM and ADB as Recovery Tools

Magnet AXIOM is a digital forensic software that is used for acquiring and analysing data from computers, mobile devices, and cloud (Magnet, 2025). It also helps in acquiring data from Android through USB using protocols such as ADB. In the case of unlocked or rooted devices, AXIOM can go further and retrieve system files, application data, messages, and metadata. It also has a strong parsing and categorizing of artifacts so that it can quickly provide insights and timeline on events.

On the other hand, ADB is a command-line tool that comes with the Android Software Development Kit (SDK) (Android, 2025). It is not a forensic tool per se but it allows for direct interaction with Android devices and can, for example, download files from the device by typing `adb pull`. Its use in forensic purposes is dependent on the settings of the device whereby USB debugging must be on, and for more advanced control, root access is usually needed.

2.5 Literature Review

Android forensic acquisition has been widely researched in the scientific and technical literature with a focus on the efficiency, shortcomings, and legal issues of various approaches. Hoog (2011) gives an outline of the Android architecture and the challenges of acquiring a forensically sound image of the device. The researcher noted that due to Android's open-source system and the variety of devices, forensic practitioners should choose methods more cautiously and based on the device and its version.

One of the main technical issues in Android forensics is the flash memory that is usually used in Android devices, known as NAND flash. According to Ahn and Lee (2021), unlike traditional hard drives, flash memory has wear-leveling and garbage collection techniques that make the data storage non-linear and overwrite the deleted data blocks frequently. This poses a problem to forensic recovery, especially when using logical acquisition techniques that involve utilizing standard APIs or file system level access. According to Hoog (2011), such methods do not allow to restore data or artifacts from unallocated space, which demonstrates the drawbacks of high-level data extraction tools in identifying traces of prior activity.

According to Wang et al. (2024), an alternative memory type is NOR flash, which allows faster random-access reads and more reliable bit-level access. While it is less common in consumer Android devices, it is found in embedded systems and may offer better forensic control due to simpler data management mechanisms. The flash translation layer (FTL) in NAND complicates physical imaging by abstracting logical block addresses from physical ones (Alahmadi and Chung, 2023). Consequently, even physical extraction might not capture the true state of user-deleted data. Obtaining a complete and forensically accurate physical image from such memory is challenging. File

system acquisition methods, especially those that use rooted ADB or commercial tools, often provide a more stable and interpretable view of data by working within the OS's structure.

Moreover, Hassan and Pantaleon (2017) go further to extend the legal and ethical analysis to more sophisticated techniques like rooting. They also point out that while rooting a device, although technically allows to gain access to the protected areas like /data/data and then copy files from there, it is dangerous because it might change some metadata or trigger system changes that may affect the integrity of the evidence. This is especially so in the newer Android versions where full-disk or file-based encryption may render rooting impossible without changing the bootloader or the secure booting process.

In his study, Kumar (2021) has looked at various tools such as Cellebrite and XRY, among others. He also discovered that commercial tools offer better formatted results and are more suitable to be used in court since they have a consistent format and record of events. But he also noted that manual tools are more versatile and can sometimes be more invasive, especially when it comes to rooted devices or devices running on older Android versions that do not have enforced encryption.

Further, Breeuwsma and Jongh (2007) also investigated the consequences of obtaining information from flash memory in the mobile phones. They include the variability of flash memory and the absence of uniformity in the devices, which complicates the process of imaging. For instance, physical images obtained by physical acquisition methods may not be an exact mirror of the live state of data because of the dynamic nature of the FTL. This supports the use of file system acquisition techniques including those that involve rooted ADB access or Magnet AXIOM, which work with the structure of the data rather than extracting it at the bit level.

Other NIST documents like the SP 800-101 Rev. 1 Richard Brothers and Wayne (2014) also offer more guidance and emphasizes on the use of validated tools and the chain of custody. NIST recognizes that the process of data recovery is stochastic and depends on the technique used in acquisition and advocates for the use of multiple techniques where possible; logical and file system for devices that cannot be acquired physically or where it is unlawful to do so.

As Android security becomes more robust with components like Trusted Execution Environments (TEE) and hardware security modules like Google's Titan M, forensic practitioners face increasing barriers. According to Junior et al. (2024), future research may focus on developing non-invasive acquisition methods that respect full-disk encryption, collaborating with OEMs for lawful access to secure modules, and building emulation or virtualization environments to analyse encrypted app containers. The field must also address challenges with cloud-stored data, cross-device sync, and AI-generated content. As digital forensics shifts toward cloud-aware, cross-platform investigations, forensic tools and methodologies must evolve to maintain evidentiary integrity and legal admissibility.

3 Method and Findings

3.1 Device Setup and Justification

The practical experiment was performed on Samsung Galaxy Note 3 (SM-N9005) with Android 5.0 (Lollipop) as its operating system. This device was chosen because it supports USB Debugging, is rooted and has an older Android version which provides a more direct access to the Android file system as opposed to the newer Android versions which have stronger full disk encryption and scoped storage. This was done before the experiment to facilitate privilege for both Magnet AXIOM and ADB extraction methods. The rationale for this practical aspect was to try to obtain as many digital evidence as possible through two different approaches: One of the commercial tools that can be used for Android device investigation is Magnet AXIOM, which employs the ADB Unlocked method. The other method is non-commercial manual executing of ADB commands with the help of Android SDK Platform-Tools. Both tools connected to the device through USB using ADB Debugging, and data collection was done in a way that could be repeated and free of errors.

3.2 Method 1: Magnet AXIOM (ADB Unlocked Acquisition)

Magnet AXIOM is a commercial forensic suite widely used in industry for mobile, computer, and cloud data acquisition and analysis. In this experiment, the AXIOM Process module was used to acquire data from the rooted Samsung device listed in **Figure 1**. Since the device was unlocked and rooted, the ADB Unlocked acquisition method was selected. This method allowed AXIOM to perform a full file system acquisition by leveraging elevated ADB access to bypass standard security restrictions. During the setup, the following image types were available:

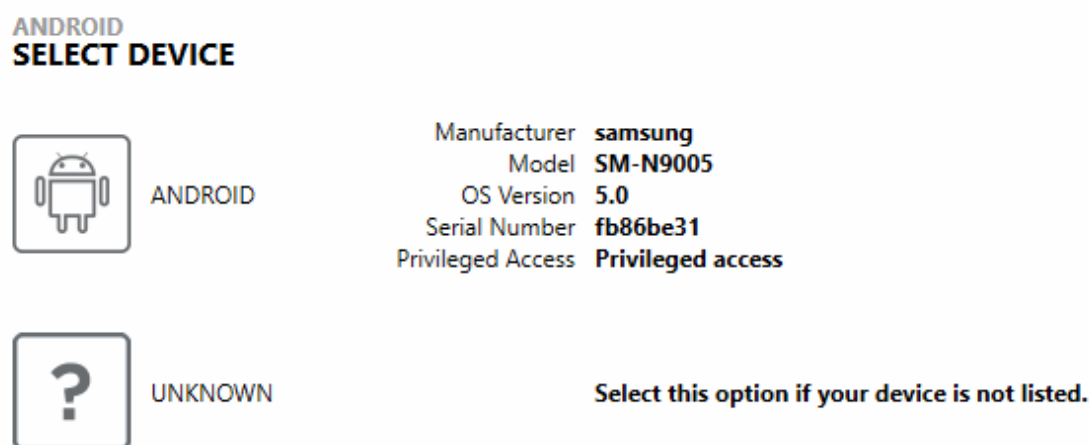


Figure 1: Listed Device with Privileged Access

- **Quick:**

Extracts data from native and third-party applications, media, and external shared storage.

- **Logical:**

Gathers data from user-accessible folders.

- **Full:**

Acquires all contents of the device's internal storage.

The Full image type was chosen to ensure a comprehensive acquisition, including protected and system-level data as shown below.

ANDROID SELECT IMAGE TYPE

Please select the type of image you want to acquire:

Full

☒ Entire contents of the device [More info](#)

Logical

☐ Entire contents of user data folders [More info](#)

Quick

☐ Native and 3rd party application data, media, and external shared storage (SD card) [More info](#)

Figure 2: Entire Device Contents (Full) Acquisition

3.2.1 Data Categories Recovered via AXIOM

i. Web Artifacts:

- Firefox bookmarks
- Chrome top sites
- Samsung Internet browser bookmarks
- Potential browser history and cached data

ALL EVIDENCE	29,173	URL	User...	Artifact type	Source	Reco...	...
REFINED RESULTS	104	http://www.w3.org/2001/XMLSchema-instance		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	
Identifiers - Device	68	http://www.w3.org/2001/XMLSchema-instance		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	
Identifiers - People	22	http://www.opengis.net/gml		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	
Social Media URLs	10	http://www.opengis.net/pidflo/1.0		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	
User Accounts	4	http://uri.etsi.org/Ingn/params/xml/simservs/xcap		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	
WEB RELATED	229	http://www.w3.org/XML/1998/namespace		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	
Android Firefox Bookmarks	1	http://ns.adobe.com/xap/1.0/		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	
Chrome Top Sites	1	http://ns.adobe.com/xap/1.0/		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	
Potential Browser Activity	222	http://www.w3.org/XML/1998/namespace		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	
Samsung Browser Bookmarks	5	http://www.google.com/transport/p2p		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	
		http://picasaweb.google.com/		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	
		http://xml.org/sax/handlers/DeclHandler		Potential Browser Activity	samsung SM-N9005 Full Image - MMCBLK0.raw - Pa...	Carving	

Figure 3: Web Artifacts

ii. Communication Data:

- Android contacts and call logs

ALL EVIDENCE	29,173				
REFINED RESULTS	104				
Identifiers - Device	68				
Identifiers - People	22				
Social Media URLs	10				
User Accounts	4				
WEB RELATED	229				
Android Firefox Bookmarks	1				
Chrome Top Sites	1				
Potential Browser Activity	222				
Samsung Browser Bookmarks	5				
COMMUNICATION	4				
Android Contacts	4				

Artifact	Key detail	Supporting detail	Additional detail
Communication Android Contacts	Display Name giffgaff voicemail	Phone Number(s) 443	Number of Times Contacted 0
Communication Android Contacts	Display Name giffgaff balance	Phone Number(s) *100#	Number of Times Contacted 0
Communication Android Contacts	Display Name giffgaff minutes	Phone Number(s) *100*7#	Number of Times Contacted 0
Communication Android Contacts	Display Name giffgaff top-up	Phone Number(s) 43430	Number of Times Contacted 0

Figure 4: Communication Data

iii. Media Files:

- User-captured photos and videos

Artifacts

MATCHING RESULTS27,237

MEDIA27,233

Pictures27,201

Videos32

APPLICATION USAGE4

MATCHING RESULTS (27,233 of 27,234)

Sort by

Small

Thumbnail view

0



1



2



3



4



5



6



7



8



9



10



11



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Figure 5: Media Files

iv. Documents:

- PDF files and plain text documents

ALL EVIDENCE	29,173				
REFINED RESULTS	104				
WEB RELATED	229				
COMMUNICATION	4				
MEDIA	27,320				
EMAIL & CALENDAR	4				
DOCUMENTS	400				
PDF Documents	15				
Text Documents	385				

Filename	Size...	Last Modified D...	Last Accessed D...	Created Date/Ti...
resources.txt	2188	08/01/2008 12:00:00 PM	08/01/2008 12:00:00 PM	12/26/2014 3:42:59 AM
CSCVersion.txt	14	08/01/2008 12:00:00 PM	08/01/2008 12:00:00 PM	12/26/2014 3:42:58 AM
apks_count_list.txt	14531	02/25/2015 1:16:42 PM	02/25/2015 1:16:42 PM	01/01/1970 12:00:00 AM
userdata_apks_count_list.txt	0	02/25/2015 1:16:42 PM	02/25/2015 1:16:42 PM	01/01/1970 12:00:00 AM
csc_apks_list.txt	0	08/08/2014 11:21:43 PM	08/08/2014 11:21:43 PM	08/08/2014 11:21:43 PM
purenandpackages.txt	1422	02/25/2015 1:16:42 PM	02/25/2015 1:16:42 PM	01/01/1970 12:00:00 AM
ringtones_count_list.txt	1526	02/25/2015 1:16:42 PM	02/25/2015 1:16:42 PM	01/01/1970 12:00:00 AM
purenandpackages_pref.txt	1449	02/25/2015 1:16:42 PM	02/25/2015 1:16:42 PM	01/01/1970 12:00:00 AM
csc_user_apks_list.txt	0	08/08/2014 11:21:43 PM	08/08/2014 11:21:43 PM	08/08/2014 11:21:43 PM
csc_ringtones_list.txt	0	08/08/2014 11:21:43 PM	08/08/2014 11:21:43 PM	08/08/2014 11:21:43 PM
hidden_apks_list.txt	0	08/08/2014 11:21:43 PM	08/08/2014 11:21:43 PM	08/08/2014 11:21:43 PM
csc_remove_apks_list.txt	0	08/08/2014 11:21:43 PM	08/08/2014 11:21:43 PM	08/08/2014 11:21:43 PM

Figure 6: Documents

v. Connected Devices:

- Previously paired Bluetooth devices

ALL EVIDENCE	29,173
REFINED RESULTS	104
WEB RELATED	229
COMMUNICATION	4
MEDIA	27,320
EMAIL & CALENDAR	4
DOCUMENTS	400
APPLICATION USAGE	566
OPERATING SYSTEM	26
CONNECTED DEVICES	48
Bluetooth Devices	48

Artifact	Key detail	Supporting detail
Connected Devices Bluetooth Devices	MAC Address 64:C5:09:50:3F:6D	Name Yalda's MacBook Pro
Connected Devices Bluetooth Devices	MAC Address 94:65:2D:42:AA:9E	Name OnePlus 5T
Connected Devices Bluetooth Devices	MAC Address 7F:46:A4:FC:20:E2	
Connected Devices Bluetooth Devices	MAC Address 72:3B:BF:0B:A0:A8	
Connected Devices Bluetooth Devices	MAC Address 50:91:75:F6:5E:0D	
Connected Devices Bluetooth Devices	MAC Address 7D:A0:12:17:FD:D6	
Connected Devices Bluetooth Devices	MAC Address 6C:96:CF:80:AE:0E	Name Shona's iPhone
Connected Devices Bluetooth Devices	MAC Address C8:14:79:20:CC:20	Name Blue van man

Figure 7: Bluetooth Devices

vi. Location and Travel:

- Wi-Fi profile metadata
- Google Maps location traces

ALL EVIDENCE	29,173
REFINED RESULTS	104
WEB RELATED	229
COMMUNICATION	4
MEDIA	27,320
EMAIL & CALENDAR	4
DOCUMENTS	400
APPLICATION USAGE	566
OPERATING SYSTEM	26
CONNECTED DEVICES	48
LOCATION & TRAVEL	14
Android Wi-Fi Profiles	1
Google Maps	13

Artifact	Key detail	Supporting detail
Location & Travel Android Wi-Fi Profiles	Network Name (SSID) Natcho_WiFi	
Location & Travel Google Maps		
Location & Travel Google Maps		
Location & Travel Google Maps	Search Query %	
Location & Travel Google Maps		
Location & Travel Google Maps	Search Query loc:	
Location & Travel Google Maps	Search Query (%.7f,%.7f)	
Location & Travel Google Maps	Search Query (	
Location & Travel Google Maps	Search Query (%f,%f)	
Location & Travel		

Figure 8: Google Maps and Wi-Fi Profiles

vii. Calendar and Email:

- Calendar entries and basic email metadata (headers and accounts)

viii. Operating System Metadata:

- Android version details
- Installed applications and package information
- Last reboot time and uptime data

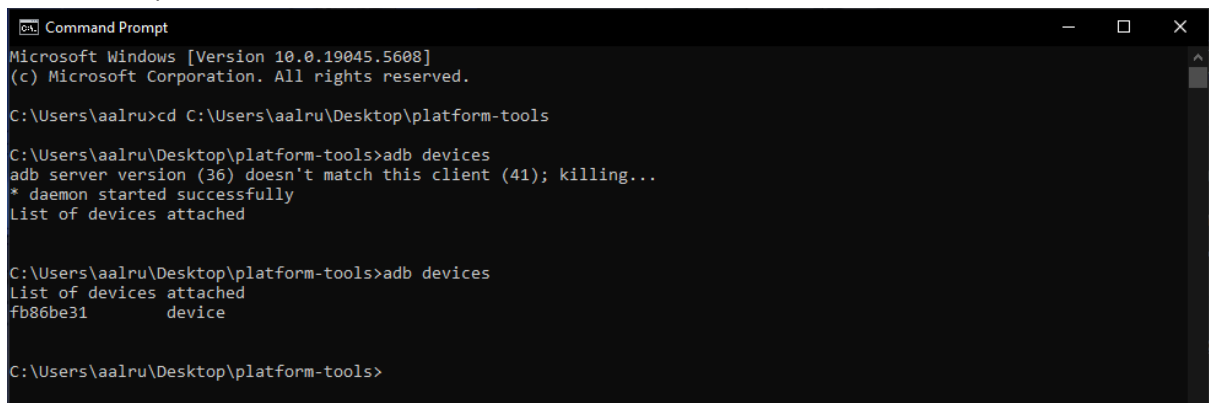
AXIOM's GUI automatically categorized these artifacts and generated structured reports, including visual timelines and filtered views. The tool also preserved hash values to maintain evidentiary integrity, an important consideration in professional forensic settings.

3.3 Method 2: ADB Manual Extraction

The second method involved using the Android Debug Bridge (ADB) from the Android SDK Platform-Tools. This is a non-commercial, command-line based toolset, enabling direct interaction with an Android device over USB when USB Debugging is enabled. Because the device was rooted, ADB was granted super user privileges, allowing access to protected directories.

The key commands used were:

- adb devices
- adb shell
- adb pull /storage/emulated/0
- adb pull /data/data



```
Command Prompt
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. All rights reserved.

C:\Users\aalru>cd C:\Users\aalru\Desktop\platform-tools

C:\Users\aalru\Desktop\platform-tools>adb devices
adb server version (36) doesn't match this client (41); killing...
* daemon started successfully
List of devices attached

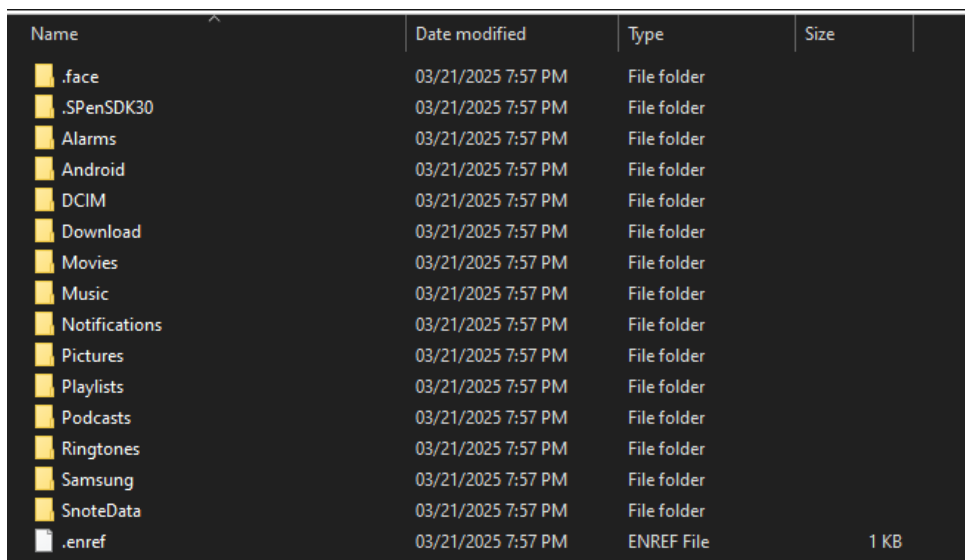
C:\Users\aalru\Desktop\platform-tools>adb devices
List of devices attached
fb86be31      device

C:\Users\aalru\Desktop\platform-tools>
```

Figure 9: adb devices command

These commands enabled to extract User directories such as /storage/emulated/0, which included:

- Alarms, Android (app data), DCIM (photos/videos), Downloads, Movies, Music, Notifications, Pictures, Podcasts, Ringtones, SNoteData, and hidden folders



Name	Date modified	Type	Size
.face	03/21/2025 7:57 PM	File folder	
.SPenSDK30	03/21/2025 7:57 PM	File folder	
Alarms	03/21/2025 7:57 PM	File folder	
Android	03/21/2025 7:57 PM	File folder	
DCIM	03/21/2025 7:57 PM	File folder	
Download	03/21/2025 7:57 PM	File folder	
Movies	03/21/2025 7:57 PM	File folder	
Music	03/21/2025 7:57 PM	File folder	
Notifications	03/21/2025 7:57 PM	File folder	
Pictures	03/21/2025 7:57 PM	File folder	
Playlists	03/21/2025 7:57 PM	File folder	
Podcasts	03/21/2025 7:57 PM	File folder	
Ringtones	03/21/2025 7:57 PM	File folder	
Samsung	03/21/2025 7:57 PM	File folder	
SnoteData	03/21/2025 7:57 PM	File folder	
.enref	03/21/2025 7:57 PM	ENREF File	1 KB

Figure 10: User Directories extracted

App-specific directories from /data/data, including:

- com.android.chrome
- org.mozilla.firefox
- com.samsung.android.messaging
- com.google.android.apps.maps

This level of access allowed manual retrieval of SQLite databases, cache files, shared preferences, and even authentication tokens, depending on app configurations.

3.3.1 Observations:

The data recovered using ADB was raw and unparsed, requiring manual examination of SQLite databases and XML files. For example, browser history in Chrome was found in the SQLite file:

/data/data/com.android.chrome/app_chrome/Default/History

Contacts and SMS data were retrievable in /data/data/com.android.providers.contacts/databases and com.android.providers.telephony/

Media and documents were directly pulled from their respective folders in internal storage.

Although ADB provided greater flexibility and control, it demanded a high level of technical skill to interpret the recovered data effectively. Furthermore, the command-line process lacked native logging, automated reporting, and artifact parsing elements that AXIOM offered by default.

3.4 Comparative Findings

The **Table 1** below summarizes the findings of both methods across major categories.

Table 1: Axiom versus ADB Pull

Category	Magnet AXIOM (Full)	ADB Pull (Manual)
Web Data	Parsed browser history/bookmarks	Raw SQLite files (manual parsing required)
Contacts/Calls	Structured contacts, call logs	Contacts.db, calllog.db extracted manually
Media	Photos, videos categorized and previewable	Full access to DCIM and Pictures folders
Documents	Parsed PDFs, text docs	Manually extracted from Download/Documents
App Data	Auto-detected installed apps + usage	Raw app folders from /data/data
System Metadata	OS version, uptime, Wi-Fi, Bluetooth	Raw system config files, logs manually parsed

Automation & Reporting	Automated reports, hashing, timelines	Manual logging and interpretation
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The two were equally efficient in extracting the forensic artifacts, although the usability and efficiency of the two were not the same. Magnet AXIOM had the features of simplicity, professional reporting, and forensic level, which were suitable for formal investigation. On the other hand, ADB offered deeper access and customization, especially when working with devices not properly supported or formats that were proprietary in nature but it was not automated and required a lot of manual work.

4 Discussion and Conclusions

4.1 Practical Implications of the Findings

The analysis of the practical comparison of Magnet AXIOM (ADB Unlocked) and manual ADB command-line extraction shows the advantages and disadvantages of both approaches in terms of digital forensics. Magnet AXIOM made it evident that it is an automated tool that is easy to use and is forensic-friendly. This is due to the fact that it was able to acquire the entire device and then parse, categorize and visualize the artifacts in a way that is easily understood by both technical and non-technical personnel (as is often the case in law enforcement or legal proceedings). Moreover, the hashing functions and the ability to generate reports for evidence also meet the evidence admissibility criterion crucial in courtroom settings.

On the other hand, the manual ADB method was found to be very flexible and technically sound especially when it came to accessing directories such as /data/data which are usually hard to access in non-rooted or non-privileged environment. The ability to extract app-specific files such as SQLite databases and cache folders gave the examiner more granular control over the forensic process. However, this method required significantly more expertise in file system structures, database forensics, and Android internals. Moreover, the lack of built-in validation mechanisms increases the risk of human error and evidentiary contamination, unless strict documentation is maintained throughout.

Both methods were effective in acquiring user-generated content and application artifacts, though AXIOM's parsing engine provided immediate insights, while ADB demanded post-processing. In real-world forensic investigations, these trade-offs must be weighed carefully, considering case objectives, time constraints, and legal requirements.

Comparison with Prior Research

The findings of this project align closely with existing literature. As highlighted by Hoog (2011), Android forensic acquisition requires a case-by-case approach due to the fragmentation of Android devices and OS versions. This experiment confirmed that older Android devices with root access can yield more comprehensive data extractions when using manual methods like ADB.

The importance of understanding flash memory behaviour, noted by Breeuwsma and Jongh (2007), was evident in the inconsistency of deleted file recovery. While AXIOM retrieved structured data effectively, neither methods reliably recovered deleted data from the test device likely due to the wear-leveling and garbage collection mechanisms in NAND flash storage. This supports prior research suggesting that logical and file system acquisition methods are limited in recovering deleted content unless physical imaging is possible which was outside the scope of this project.

Hassan and Pantaleon (2017) raised concerns about rooting in forensic contexts, particularly regarding its legal defensibility. In this experiment, rooting provided clear technical benefits, but in real-world cases, it would present legal and ethical challenges especially in jurisdictions where modifying a device before acquisition could be interpreted as tampering.

4.2 Improvements and Recommendations

Combining the strengths of commercial tools and manual methods can produce better results. For instance, ADB can be used to access unsupported or encrypted app directories, while AXIOM can handle broader acquisition and structured reporting.

Where rooting is not legally permissible, investigators should explore exploit-based full-disk acquisition techniques (for example, through custom recovery environments or vendor-specific solutions), provided they maintain forensic soundness.

Investigators using manual tools like ADB must maintain detailed logs and practice strict procedural control. Creating scripts for standard extractions and documenting each step can enhance reproducibility and reduce risk.

Whether using ADB or AXIOM, practitioners should regularly test their tools on known datasets to ensure consistent and reliable behaviour. This aligns with NIST's emphasis on tool validation for forensic defensibility.

For investigations requiring deleted data recovery, future work should incorporate physical imaging methods and research into carving techniques specific to flash memory's structure.

4.3 Final Conclusions

This project successfully evaluated and compared two data acquisition methods for Android forensic investigation; Magnet AXIOM using ADB Unlocked acquisition and manual extraction via ADB commands. The results demonstrate that both approaches are capable of recovering critical forensic artifacts, but they serve different roles in the investigative process.

Magnet AXIOM provides a structured, legally defensible workflow ideal for professional environments, while ADB offers a powerful, flexible alternative suited for technically proficient investigators dealing with non-standard or older devices. However, ADB's benefits come at the cost of increased complexity and potential legal or ethical concerns if used inappropriately.

Ultimately, the choice of tool should depend on the investigative goals, the technical condition of the device, and the legal constraints under which the examiner operates. In practice, a balanced and context-aware approach supported by continuous professional development and ethical awareness, essential for conducting effective and defensible digital forensic investigations.

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